Desaboina Sri Sathwik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.26
Sri Chaitanya junior college, khammam	Class XII (TGBIE)	2024	96.90%
Sri Chaitanya olympiad school, khammam	Class X (TGBSE)	2022	10.00%

PROJECTS

Autonomous Mobile Robot using ROS 2 & LiDAR SLAM | Github link | *Ongoing Project*

Technologies: ROS 2, SLAM Toolbox, LiDAR, Gazebo, URDF/Xacro, RViz

- Developing a four-wheel autonomous mobile robot in **ROS 2**, including robot modeling using **URDF/Xacro** and simulation in **Gazebo**.
- Integrated a **LiDAR sensor** and implemented **2D SLAM using SLAM Toolbox** for real-time environment mapping and robot localization.
- Configured and tuned SLAM parameters and validated the generated map and TF relationships using **RViz** and ROS 2 tools.
- Developing the project incrementally toward autonomous navigation, with **Nav2-based path planning, localization, and obstacle avoidance** planned as the next stage.
- **Future work:** Integrate a robotic manipulator and develop **autonomous pick-and-place** capabilities, combining mobile navigation with robotic arm manipulation.

Deep Learning Architectures from Scratch | Github Link

- Implemented CNN components from scratch in **NumPy** and trained image classification models in **PyTorch** on CIFAR-10 using multiple optimization algorithms.
- Built **Vanilla RNN, LSTM, and Transformer Encoder** from scratch for sentiment classification, implementing BPTT, attention mechanisms, and sequence modeling techniques.
- Conducted experiments involving dropout, batch normalization, t-SNE visualization, optimizer comparisons, and sequence-length analysis to evaluate model performance

Mini Machine Learning Library | Github Link

- Developed a mini deep learning library from scratch in Python without relying on high-level deep learning frameworks.
- Implemented a custom automatic differentiation (autograd) engine supporting forward and backward propagation.
- Built neural network components including layers, activation functions, optimizers, and training loops.
- Implemented classical machine learning algorithms such as Logistic Regression and Naive Bayes, along with data loading and model evaluation utilities.
- Validated the library through experiments on spam classification and visualized training loss and accuracy.

FitLLM -- AI-Powered Fitness & Diet Plan Generator | Github Link

- Built an AI fitness coach by fine-tuning an open-source language model to generate personalized workout and diet plans from a user's profile.
- Engineered a custom instruction-tuning dataset from real exercise and gym-member records, computing nutrition targets using the Mifflin-St Jeor formula.
- Fine-tuned the base model using LoRA (Low-Rank Adaptation), updating a small fraction of total parameters instead of the full model.
- Loaded and ran the model in 4-bit quantization using Hugging Face Transformers, PEFT, and TRL for memory-efficient training on a single GPU.
- Evaluated the fine-tuned model against the base model on held-out profiles to verify the fine-tuning improved plan quality and relevance.
- Deployed the final model as an interactive web demo using Gradio.

Fair Bayesian Active Learning | Github Link

- Developed a fairness-aware Bayesian Active Learning framework by integrating Bayesian uncertainty estimation with fairness-aware sample acquisition strategies.

- Implemented and evaluated Bayesian variants of FAL- Aggregate, FAL-Nested, and FAL-Nested-Append using **PyTorch** on the **Give Me Some Credit** dataset.
- Evaluated model performance using **Accuracy, AUC-ROC, F1-score, Demographic Parity Gap, Equalized Odds (TPR Gap),** and **Expected Calibration Error (ECE)** to analyze the fairness-accuracy trade-off.
- Did this project as part of the **AI3013: AI for Humanity** course.

RELEVANT COURSES

Core Artificial Intelligence & Machine Learning: Introduction to Modern AI, Programming for AI, Foundations of Machine Learning, Deep Learning, AI for Humanity

Core Engineering: Linear Systems & Signal Processing, Control Systems

Mathematics for AI & Machine Learning: Probability & Random Variables, Optimization I & II, Numerical Analysis, Matrix Theory, Vector Calculus, Differential Equations, Introduction to Metric Spaces

Computer Science : Programming, Data Structures & Applications, Algorithms, Database Management Systems (DBMS-I), Discrete Mathematics for Computer Science

SKILLS

Robotics : ROS 2, Nav2, SLAM Toolbox, Gazebo

Deep Learning: Neural Networks (ANN, CNN), Transformers, PyTorch, Transfer Learning, LLM Fine-tuning, Hugging Face Transformers

Programming Languages: Python, C, C++

Libraries & Frameworks: PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn

Tools & Platforms: Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Linux

Databases: SQL

EXTRACURRICULAR

- National Service Scheme(NSS) - volunteer

Volunteer, OCS, IIT Hyderabad

- Volunteered in organizing and supporting career-related events and placement activities conducted by the Office of Career Services.